2.1

(a)

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	2	4
2	2	1	0	0
1	3	0	1	0
-1	1	-2	-1	2
0	0	-1	-2	2
$\Sigma x_i =$ 5	$\Sigma y_i =$ 10	$\Sigma(x_i-\bar{x}) =$ 0	$\Sigma(y_i-\bar{y}) =$ 0	$\Sigma(x_i-\bar{x})(y_i-\bar{y})$ 8

$$\bar{x} = \frac{3+2+1+(-1)+0}{5} = 1$$

$$\bar{y} = \frac{4+2+3+1+0}{5} = 2$$

(b)

$$b_2 = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = \frac{8}{4+1+4+1} = \frac{8}{10} = \frac{4}{5}$$

$$b_1 = \bar{y} - b_2\bar{x} = 2 - \frac{4}{5}\times 1 = \frac{6}{5}$$

$\begin{cases} b_2 = \frac{4}{5}, & \text{means slope of the fitted regression} \\ b_1 = \frac{6}{5}, & \text{means intercept of the fitted regression} \end{cases}$

(c)

$$\sum_{i=1}^{5} x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15, \quad \sum_{i=1}^{5} x_i y_i = 12 + 4 + 3 + (-1) + 0 = 18$$

① $\Sigma(x_i-\bar{x})^2 = 4+1+4+1 = 10$, $\Sigma x_i^2 - N\bar{x}^2 = 15 - 5\times 1^2 = 10$ $\therefore \Sigma(x_i-\bar{x})^2 = \Sigma x_i^2 - N\bar{x}^2$

② $\Sigma(x_i-\bar{x})(y_i-\bar{y}) = 8$, $\Sigma x_i y_i - N\bar{x}\bar{y} = 18 - 5\times 1\times 2 = 8$ $\therefore \Sigma(x_i-\bar{x})(y_i-\bar{y}) = \Sigma x_i y_i - N\bar{x}\bar{y}$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3,6	0,4	0,16	1,2
2	2	2,8	-0,8	0,64	-1,6
1	3	2	1	1	1
-1	1	0,4	0,6	0,36	-0,6
0	0	1,2	-1,2	1,44	0
$\Sigma x_i =$ 5	$\Sigma y_i =$ 10	$\Sigma\hat{y}_i =$ 10	$\Sigma\hat{e}_i =$ 0	$\Sigma\hat{e}_i^2 =$ 3,6	$\Sigma x_i\hat{e}_i =$ 0

$\hat{y}_i = \frac{6}{5} + \frac{4}{5}x_i$

$\frac{6}{5} + \frac{4}{5}\times 3 = 3,6$, $\frac{6}{5} + \frac{4}{5}\times 2 = 2,8$

$\frac{6}{5} + \frac{4}{5}\times 1 = 2$, $\frac{6}{5} + \frac{4}{5}\times(-1) = 0,4$

$\frac{6}{5} + \frac{4}{5}\times 0 = 1,2$

$$S_y^2 = \frac{\sum_{i=1}^{N}(y_i-\bar{y})^2}{N-1} = \frac{4+1+1+4}{4} = 2,4$$

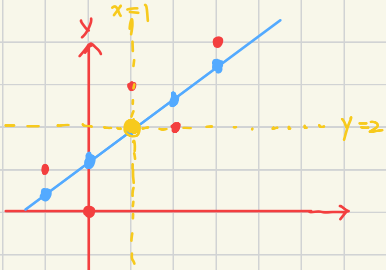
$$S_x^2 = \frac{\sum_{i=1}^{N}(x_i-\bar{x})^2}{N-1} = \frac{4+1+1+4}{4} = 2,4$$

$$S_{xy} = \frac{\sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})}{N-1} = \frac{8}{4} = 2, \quad r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{\sqrt{2,4\times 2,4}} = 0,8$$

$$CV_s = 100\left(\frac{S_x}{\bar{x}}\right) = 100\times\frac{\sqrt{2,4}}{1} \doteqdot 158,11$$

Ans: $S_y^2 = S_x^2 = 2,4$, $S_{xy} = 2$, $r_{xy} = 0,8$, $CV_s = 158,11$

(e)



(f) 用黃色標於上圖，位於 (1,2)

(g) $b_1 + b_2\bar{x} = \frac{6}{5} + \frac{4}{5} \times 1 = 2 = \bar{y}$

(h) $\bar{\hat{y}} = \sum \hat{y}_i / N = 10/5 = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum (y_i - \hat{y})^2}{N-2} = \frac{0.4^2 + (-0.8)^2 + 1^2 + 0.6^2 + (-1.2)^2}{3} = \frac{3.6}{3} = \underline{1.2}$ #

(j) $\widehat{Var}(b_2 | x) = \frac{\hat{\sigma}^2}{S_{xx}} = \frac{1.2}{10} = \underline{0.12}$ #

$se(b_2) = \sqrt{0.12} = \underline{0.34641}$ #